OnEvoMemory: Evolving Memory through Online Robot Rollouts for Pretrained Robot Policies

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Abstract. Long-horizon robot manipulation requires policies to track completed subtasks and critical interaction events. However, existing memory mechanisms heavily rely on external models or predefined update rules. To address this, we propose **OnEvoMemory**, a value-guided memory module for pretrained robot policies. It maintains recent context, high-value experiences, and salient transitions, while learning which experiences should be retained from trajectory outcomes. Offline demonstrations initialize the memory prior, whereas successful and unsuccessful online rollouts refine memory selection, helping the policy recognize task-stage transitions and avoid repeating completed subtasks. Experiments on long-horizon manipulation benchmarks show that OnEvoMemory improves the performance of the base VLA policy through both offline initialization and online memory evolution.

Keywords: Long-horizon manipulation · Vision-language-action models

1 Introduction

Embodied policies, such as vision-language-action (VLA) models [4] and diffusion policies [2], continue to face substantial challenges in long-horizon task execution and often benefit from additional memory mechanisms [5, 8, 9]. Existing approaches commonly augment policies with fixed temporal windows, similarity-based retrieval, or manually designed writing rules [5, 8, 13–15]. However, these methods typically predetermine what information should be retained, making it difficult to adapt the focus of memory to the requirements of different tasks based on actual interaction outcomes. In long-horizon manipulation, certain events—such as establishing a successful grasp, losing contact with an object, or entering a new task stage—can have a considerable influence on subsequent decisions [14, 15]. This raises a central question: how can a model learn what is worth remembering?

To address this question, as illustrated in Fig. 1, we introduce **OnEvoMemory**, a value-guided memory system that is initialized from offline experience and continuously evolves through online robot rollouts.

Our main contributions are summarized as follows:

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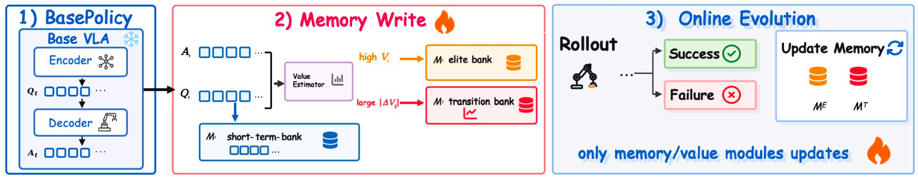

Fig. 1: Overview of OnEvoMemory. A frozen base VLA produces action-query representations, which are organized into short-term, elite, and transition memories through value-guided writing. Successful and unsuccessful robot rollouts further update only the value and memory modules, enabling online memory evolution without modifying the pretrained policy.

- We introduce a hierarchical memory architecture consisting of a short-term raw buffer, an elite experience bank, and a transition bank, which capture recent context, high-value experiences, and salient value transitions, respectively.
- We develop a learnable, value-guided memory-writing mechanism that identifies task-relevant experiences without relying on fixed temporal windows or manually defined writing rules.
- We propose an online memory-evolution approach that uses successful and unsuccessful rollouts to refine the memory mechanism while keeping the underlying policy frozen.

2 Related Work

Recent memory-augmented robot policies mainly differ in how they select and maintain historical information [5, 8, 9, 12]. One line of work relies on external vision–language models to identify task-relevant history. For example, BPP uses an off-the-shelf VLM to detect semantic keyframes, while MemER employs a high-level VLM to select previous observations and produce subtask instructions for a low-level policy [7,10]. Although these methods benefit from strong semantic priors, their memory selectors are decoupled from the underlying robot policy, making it difficult to continually adapt memory selection from online interaction outcomes.

Another line of work adopts predefined memory structures or update heuristics [14,15]. MemoryWAM, for instance, combines a sliding window of recent observations, task-boundary anchor frames, and compact gist tokens to efficiently preserve long-range context [13]. Such designs provide effective and computationally efficient history representations, but what should be retained is largely determined by the initial memory mechanism. In contrast, OnEvoMemory uses rollout outcomes to continually refine a value-guided memory writer, enabling memory selection to evolve online while keeping the pretrained policy frozen.

3 Method

Overview. We propose **OnEvoMemory**, a value-guided memory mechanism initialized from offline demonstrations and continuously adapted through online robot rollouts. We instantiate it on a token-based VLA policy, while the proposed mechanism can also be applied to policies that expose intermediate action representations.

Given an observation $o_t = (I_t^{1:V}, s_t, \ell)$, where $I_t^{1:V}$, s_t, and ℓ denote multi-view images, proprioceptive states, and a language instruction, respectively, the base policy produces action-query representations $Q_t = E_\theta(o_t) \in \mathbb{R}^{H\times d}$, where H is the action horizon.

OnEvoMemory maintains an episode-local memory $\mathcal{M}_t = \{\mathcal{M}_t^S, \mathcal{M}_t^E, \mathcal{M}_t^T\}$, consisting of a short-term raw buffer, an elite experience bank, and a transition bank. The current action queries retrieve historical representations from these components to form $C_t = \mathrm{Retrieve}(Q_t, \mathcal{M}_t)$, which is incorporated through gated cross-attention as $\widetilde{Q}_t = Q_t + g_t \,\mathrm{MHA}(Q_t, C_t, C_t)$. The base action decoder then predicts an action chunk $\hat{A}_t = D_\theta(\widetilde{Q}_t)$. Memory is always read before the current experience is written, preventing the current state from being immediately retrieved within the same decision step.

3.1 Value-Guided Hierarchical Memory

OnEvoMemory represents interaction history using three complementary memory components. The short-term buffer $\mathcal{M}^S$ stores recent action-query representations in a fixed-capacity FIFO queue, preserving local visual, linguistic, and proprioceptive context. The elite experience bank $\mathcal{M}^E$ retains high-value state–action experiences, such as stable grasps and successfully completed intermediate stages, while the transition bank $\mathcal{M}^T$ preserves salient changes in trajectory quality, including task progress, contact loss, failure, and recovery events.

Memory writing is guided by an action-conditioned value estimator $(V_t, k_t, v_t) = F_\phi(\widetilde{Q}_t, A_t)$, where V_t provides a trajectory-outcome-aligned writing score, and k_t and v_t denote the retrieval key and stored representation, respectively. High-value experiences are written to $\mathcal{M}^E$, whereas experiences with large temporal value changes $|V_t - V_{t-1}|$ are written to $\mathcal{M}^T$. This enables the system to identify useful experiences and salient transitions without relying on fixed temporal windows or manually specified event rules.

During retrieval, each long-term bank returns both semantically relevant and recently written experiences, following $\mathcal{R}_t^b = \mathrm{TopK}_{\mathrm{sim}}(Q_t, \mathcal{M}_t^b) \cup \mathrm{RecentK}(\mathcal{M}_t^b)$, where $b \in \{E, T\}$. The retrieved representations are combined with the short-term buffer to form the memory context C_t.

3.2 Offline Initialization and Online Evolution

During offline initialization, expert demonstrations are replayed in temporal order, such that earlier experiences populate the memory banks and subsequent observations retrieve the accumulated memory for action prediction. Action supervision and trajectory outcomes jointly initialize the memory prior and train the memory reading, writing, and injection modules.

During online rollouts, both successful and unsuccessful trajectories provide new outcome supervision, enabling the value estimator and memory writer to continually revise which experiences should be retained and reused. The pretrained vision–language backbone and action policy remain frozen throughout online adaptation, while only the value estimator and memory-related modules are updated. OnEvoMemory therefore evolves its memory selection and utilization while preserving the manipulation capability of the pretrained policy.

4 Experiments

Experimental setup. We instantiate OnEvoMemory on QwenOFT [3, 11] as the base VLA policy and evaluate it on the ten tasks of LiberoLong-10 [6] and two long-horizon tasks from RMBench [1]: SwapBlocks and SwapT. We compare three settings: the original base policy, the policy augmented with offline-initialized memory, and the same system after one round of online memory evolution. During online adaptation, we collect 20 rollout trajectories per task on LiberoLong-10 and 10 trajectories per task on RMBench. The pretrained QwenOFT policy remains frozen, and only the value estimator and memory-related modules are updated.

Table 1: Success rates (%). Offline initializes memory from demonstrations, while Online further updates it with one round of robot rollouts.

Benchmark / Task	Base VLA	+ Offline	+ Online
LiberoLong-10 Avg.	86.2	88.6	**90.2**
RMBench SwapBlocks	0	10.0	**14.0**
RMBench SwapT	0	8.0	**10.0**

Results and analysis. As shown in Table 1, offline memory initialization improves the average success rate on LiberoLong-10 from 86.2% to 88.6%, and one round of online memory evolution further raises it to 90.2%. On the more challenging RMBench tasks, the base policy achieves 0% success on both SwapBlocks and SwapT. Offline memory increases the success rates to 10% and 8%, respectively, which are further improved to 14% and 10% after online adaptation.

Qualitatively, memory is particularly helpful around task-stage transitions. By retaining high-value experiences associated with completed subtasks, the policy is less likely to lose track of its current progress and repeat actions corresponding to stages that have already been completed. The additional gains from online adaptation may be attributed to the unsuccessful rollouts collected during interaction: these negative examples provide corrective supervision for both the value estimator and the memory-writing mechanism, helping the system revise overestimated experiences and retain events that are more informative for subsequent decisions. While the improvements remain moderate and the RMBench success rates are still limited, the results suggest that offline memory provides a useful initialization and that rollout-based updates can further refine memory selection without modifying the pretrained policy.

5 Conclusion

We studied how pretrained robot policies can learn what to remember for long-horizon manipulation. OnEvoMemory combines offline memory initialization with rollout-driven online refinement, helping the policy preserve task progress and avoid repeating completed subtasks. Experiments suggest that memory selection can be improved from both successful and unsuccessful experience while keeping the base policy frozen.

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